

PH.D. QUALIFYING EXAM 2020: STATISTICAL INFERENCE

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You can take the exam at any location where you have the facility to scan or take pictures of your answers and convert them into pdf files. You are allowed to use any material of your choice prepared before the time of the exam, but use of online materials or consulting anyone else during the exam is not be permitted. If you have any questions during the exam, please email Professor Zhiqiang Tan (ztan@stat.rutgers.edu) for problem 1 or Professor David Tyler (dtyler@stat.rutgers.edu) for problem 2.

Please make sure to **handwrite** your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. **Typed answers are not permitted.**

Create two pdf files for your answers - one for each problem. Both the pdf files containing your answers should be emailed to td370@stat.rutgers.edu and hcrane@stat.rutgers.edu by 2:15 PM. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Good luck!

Problem 1

Suppose that $(Y_1, \dots, Y_n)$ are observed, where $Y_i = \max(0, X_i)$ for $i = 1, \dots, n$ and $(X_1, \dots, X_n)$ are independent and identically distributed as $N(\mu^*, \sigma^{*2})$, where (μ^*, σ^*) are unknown. For $i = 1, \dots, n$, it is useful to define the binary variable

$$R_i = \begin{cases} 1, & \text{if } Y_i > 0, \\ 0, & \text{if } Y_i = 0. \end{cases}$$

Denote by $\Phi(\cdot)$ the cumulative distribution function (CDF) of $N(0, 1)$. For the following questions, denote by P_θ the distribution $N(\mu, \sigma^2)$, with $\theta = (\beta, \sigma)$ and $\beta = \mu/\sigma$. Moreover, let $\theta^* = (\beta^*, \sigma^*)$.

i) Define an estimator $\hat{\theta}_1 = (\hat{\beta}_1, \hat{\sigma}_1)$ as a solution to the simultaneous equations

$$E_\theta(R_1) = \frac{1}{n} \sum_{i=1}^n R_i,$$

$$E_\theta(R_1 Y_1) = \frac{1}{n} \sum_{i=1}^n R_i Y_i,$$

where $E_\theta(\cdot)$ denotes the expectation taken with respect to P_θ . Find closed-form expressions for $(\hat{\beta}_1, \hat{\sigma}_1)$. You may use $\Phi(\cdot)$ and its inverse function.

ii) Show that $(\hat{\beta}_1, \hat{\sigma}_1)$ are consistent estimators of (β^*, σ^*) .

iii) Show that $\sqrt{n}(\hat{\theta}_1 - \theta^*) \rightarrow_{\mathcal{D}} N(0, \tau_1)$ as $n \rightarrow \infty$ for some variance τ_1 . Find a closed-form expression for τ_1 . You may use $\Phi(\cdot)$ and its inverse function.

iv) Write down the log-likelihood function $\ell(\theta)$, and compute $E_\theta\{\frac{1}{n} \frac{\partial \ell(\theta)}{\partial \theta}\}$ and $n \text{var}_\theta\{\frac{1}{n} \frac{\partial \ell(\theta)}{\partial \theta}\}$ in closed-form expressions, where $\text{var}_\theta(\cdot)$ denotes the variance taken with respect to P_θ .

v) The maximum likelihood estimator of θ , denoted as $\hat{\theta}_2 = (\hat{\beta}_2, \hat{\sigma}_2)$, can be shown to be consistent and $\sqrt{n}(\hat{\theta}_2 - \theta^*) \rightarrow_{\mathcal{D}} N(0, \tau_2)$ as $n \rightarrow \infty$ for some variance τ_2 . (You are not asked to prove this.) Find a closed-form expression for τ_2 .

vi) Show that $\tau_1 \geq \tau_2$. Give a direct proof, instead of invoking the asymptotic efficiency of maximum likelihood estimation.

Problem 2

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be a random sample from a multivariate $Normal_q(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution. Consider the model $\boldsymbol{\mu} = \eta \mathbf{a}$ and $\boldsymbol{\Sigma} = \alpha \mathbf{I} + \beta \mathbf{a} \mathbf{a}'$, where $\mathbf{a} \in \mathbb{R}^q$ is a fixed vector with norm one, i.e. $\mathbf{a}' \mathbf{a} = 1$, and with η, α , and β being unknown real parameters.

- (a) Find necessary and sufficient conditions on (α, β) for $\boldsymbol{\Sigma} > \mathbf{0}$, i.e. for $\boldsymbol{\Sigma}$ to be positive definite.
- (b) Find the eigenvalues of $\boldsymbol{\Sigma}$, their corresponding eigenspaces and eigenprojections, and the trace of the eigenprojections.
- (c) Find the maximum likelihood estimator $\hat{\boldsymbol{\theta}} = (\hat{\eta}, \hat{\alpha}, \hat{\beta})$ of $\boldsymbol{\theta} = (\eta, \alpha, \beta)$. *Clearly justify why it is the mle.*
- (d) Find the limiting distribution of the $\sqrt{n}(\hat{\alpha} + \hat{\beta} - \alpha - \beta)$.
- (e) Find the expected value of $\hat{\boldsymbol{\theta}}$. Use this result to find minimum variance unbiased estimators of η , of α and of β . *Clearly justify your answers.*